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Application	Use of the Existing Swimming Pool Building for Commercial Operations (Retrospective Application)
Planning Ref:	25/0574/RCU
Document Ref:	22387/TM/001
Date:	12th November 2025

TECHNICAL MEMORANDUM – REVIEW OF IMPACT ACOUSTICS NOISE IMPACT ASSESSMENT REPORT REF IMP7903-2 DATED SEPTEMBER 2025 AND REBUTTAL LETTER IN RESPONSE TO LOCAL AUTHORITY DECISION TO REFUSE PLANNING PERMISSION

ES Acoustics Ltd have been instructed by the owner/occupiers of 28 and 32 Allandale Avenue to review the Noise Impact Assessment Report prepared by Impact Acoustics (ref IMP7903-2), and the rebuttal letter produced by Impact Acoustics in response to the Officer's decision to recommend a refusal of planning permission.

This Technical Memorandum provides a review of the report from a technical perspective on the following pages.

In summary, the key issues with the report are as follows:

1. **Incorrect Application of Standards:** The report applies BS 4142, which is intended for industrial and commercial noise sources. Swimming lessons are a recreational activity involving people, which are explicitly excluded from the scope of BS 4142. Therefore, the chosen methodology is fundamentally inappropriate, invalidating any conclusions about swimming activity noise.

2. **Assessment Conditions Not Representative of Actual Operation:** The assessment assumes all doors and skylights of the swimming pool are closed, despite resident observations that they are always open during lessons. As a result, the report fails to reflect realistic operating conditions, significantly undermining its reliability.

3. **Calculation and Data Transparency Issues:**

    a. **Lack of construction details:** Section 6 provides sound reduction ("R") values for walls, roof and windows, without specifying construction build-ups, materials or data sources. This introduces significant uncertainty in the accuracy of the noise breakout calculations.

    b. **No verification of results:** key modelling inputs (distances, line of sight, etc) are not provided, meaning that results cannot be replicated or verified.

4. **Flawed / Missing Methodology for Swimming Lesson Assessment:** no clear explanation of how the swimming lesson noise levels were assessed. The report simply compares a 1-hour L_{Aeq} to background noise without citing an appropriate standard or explaining why a 2 dB difference equates to "low impact". The conclusion of "low impact" is therefore unsupported and arbitrary.

5. **Misuse of BS 8233:2014:** BS 8233 is cited for internal noise criteria, but it is meant for steady external sources (e.g. traffic), not variable, intermittent sound like human activity while swimming, or shouted instructions given by instructors to students. The report applies BS 8233 without suitable justification or discussion, making the conclusions technically invalid.

6. **L_{Amax} Assessment Deficiencies:** The report references WHO night-time criteria but fails to justify how these are relevant to daytime operations. No discussion is provided of the number or frequency of noise events expected during a typical operational day. The methodology lacks transparency, with no details on how levels were calculated. If open skylights were considered as they should have been, then predicted L_{Amax} levels exceed the report's own stated limits, contradicting its conclusions.

7. **No Assessment to Gardens:** The report only assesses noise at façades, omitting external amenity areas (gardens), which are critical for a residential noise impact assessment.

8. **Professional and Procedural Shortcomings:** The report is poorly structured, lacks justification and transparency, and does not enable independent verification. Responses to the Environmental Health Officer's (EHO) comments are defensive rather than technical, showing a misunderstanding of planning noise principles. The rebuttal wrongly expects the EHO to provide counter-data rather than demonstrating robustness in their own assessment.

In summary, the reviewed report is not fit for purpose as a planning noise impact assessment.

It applies incorrect standards to the type of noise source, fails to model real operating conditions, lacks transparency and justification in its data and calculations, and draws unsupported conclusions about the impact of swimming lessons. As a result, the conclusions regarding "low impact" cannot be considered reliable or defensible.

Should you have any questions or comments with regards to the contents of this technical memorandum, please contact me directly at the details below.

Yours Sincerely

Daniel Green MIOA
Director & Principal Acoustic Consultant
[redacted]

1 REPORT REVIEW

1.1 Report Title

The Report is titled 'Industrial Noise Impact Assessment Swimming Lessons & AC Unit'.

A swimming pool is considered a leisure or community facility and is not an industrial source. The intent of the report is therefore incorrect from the title alone, before getting to the content.

1.2 Report Methodology

The Report methodology follows the guidance presented within British Standard 4142:2014+A1:2019 titled 'Methods for rating and assessing industrial and commercial sound'. The scope of this standard covers the following:

a) *sound from industrial and manufacturing processes;*

b) *sound from fixed installations which comprise mechanical and electrical plant and equipment;*

c) *sound from the loading and unloading of goods and materials at industrial and/or commercial premises; and*

d) *sound from mobile plant and vehicles that is an intrinsic part of the overall sound emanating from premises or processes, such as that from forklift trucks, or that from train or ship movements on or around an industrial and/or commercial site.*

The scope notes that the standard is not intended to be applied to the rating and assessment of sound from:

a) *recreational activities, including all forms of motorsport;*

b) *music and other entertainment;*

c) *shooting grounds;*

d) *construction and demolition;*

e) *domestic animals;*

f) *people;*

g) *public address systems for speech; and*

h) *other sources falling within the scope of other standards or guidance.*

It is the view of ES Acoustics that swimming lessons would be considered both a 'recreational activity' and include sound from 'people'. Therefore, it is not appropriate to use British Standard 4142:2014+A1:2019 for the assessment of noise associated with swimming activity.

For this reason alone, the Report cannot possibly draw accurate conclusions on noise associated with the swimming pool use.

Section 4.2 references "car movements". BS 4142 is also inappropriate for the assessment of vehicular noise on the public highway. While it can be used when assessing "vehicles that are an intrinsic part of the overall sound emanating from premises or processes", the assessment should only consider these vehicles when operating on the application site itself. In this case, cars would be using and parking on the public highway, and the use of BS 4142 would not be appropriate.

Section 4.3 is titled "Internal Activity" and discusses doors and windows of the swimming pool being closed, and the assessment being based on this condition. This is a fundamental flaw in the assessment, as while noise may be sufficiently contained when doors and windows are sealed shut, this is not the condition under which the site has been operating. Residents have repeatedly observed skylights being open during swimming lessons, as well as, on occasion, the doors facing one of the residents are sometimes left open. The assessment does not, therefore, consider the actual operating conditions of the swimming pool use.

1.3 Report Calculations (Section 6 of the Report)

Section 6 is titled "Plant Calculations". This is confusing and unclear, given Section 6.2 refers to noise breakout, which is assumed to be in relation to noise associated with the swimming pool activity, which is not "plant".

Section 6.2 of the Report presents tables of noise breakout from the swimming pool roof, rear wall and side wall & windows combined. Each table presents an 'R' reduction value of the element in question. No detail is provided as to what the construction elements are, nor how the sound reduction values at each single octave frequency band have been derived. Given this lack of critical information, there is a high level of uncertainty as to whether these calculations are accurate. Either:

1. An investigative sound insulation test could have been conducted on site to measure the actual performance of the swimming pool's external building fabric, or
2. A clear breakdown of the construction elements could have been provided, with either calculations of the expected sound insulation values of the constructions with suitable software, or reference examples of similar constructions with suitable literature.

As noted previously, the fundamental error here is that the calculations do not consider the skylights of the swimming pool to be open, therefore meaning that the resultant calculated façade levels shown in Section 6.4.2 are inaccurate.

1.4 BS 4142 Assessment (Section 7 of the Report)

Section 7 is titled "BS4142:2014 +A1:2019 Noise Assessment". At no point in the title is it clear that this section only assesses noise from the Fujitsu AOYG14LMCA unit. It is not until arriving at Section 8 that it is apparent that the swimming pool activity is not included. This is a consistent theme throughout the report, whereby the reader is left second-guessing what the intent of the author is.

Section 7.7.1 presents an "Assessments". While it is understood that this has been conducted in CADNA A noise modelling software, no information is provided as to the distance between the units and the receptors, whether the unit is in line of sight, etc.

While the outcome of the assessment may be correct for this aspect, there is no way of verifying this, which is not appropriate for a technical report. It is expected that a technical report should provide sufficient information such that someone could repeat the process and obtain the same results. Too much information is hidden from the reader, and it can therefore not be verified as to whether this is accurate or not.

It should also be noted that a BS 4142 assessment must consider the context in which the sound occurs before the conclusions can be drawn. This assessment completely ignores this part of the assessment process. Given that the Rating Level is 16 dB below the background sound level, it is unlikely that this would change the outcome of this part of the assessment. However, due to the lack of information and the lack of process, the conclusions drawn cannot be considered reliable.

1.5 Swimming Lessons (Section 8 of the Report)

No actual methodology is presented for the assessment of swimming lessons. It is assumed that the report is intended to follow the methodology of BS 4142, given that the cover page of the report references this British Standard, but at no point is this clarified.

Section 8.2 is titled "Assessments". Again, no information is provided, other than the questionable detail provided in Section 6, as to how the noise level at the closest receptor has been calculated. The assessment provides a crude comparison of an $L_{Aeq,\ 1\text{-hour}}$ level vs a background sound level, with no discussion or justification. It is therefore impossible for the reader to know how the 'assessment conclusion' of 'low impact' has been drawn. Why does the author consider a level of -2 dB below background to be a low impact? What standard or assessment methodology is used to draw this conclusion?

The statement below the assessment table states, *"it can be seen by the above summary that the swimming lesson have returned an assessment conclusion of 'Low Impact'".* Firstly, the swimming lesson itself cannot return an assessment conclusion – this is the job of the report author. Secondly, given that it is not clear what assessment methodology is being used, and no justifications are being provided, it is the view of ES Acoustics that this conclusion cannot "be seen".

Section 8.3 of the report is titled "Partial Open Windows", and notes that a partially open window offers an attenuation value of 15 dB. What partially open window is this referring to? At every section of the Report, it is the job of the reader to guess or somehow interpret the intent of the author. It is assumed that the author is referring to the window of the receptor property, given that Section 8.3.1 goes on to reference BS 8233:2014, which recommends internal ambient noise levels for dwellings. What is not discussed is that the values presented in BS 8233:2014 are for "for steady external noise sources", such as road traffic, etc. Noise associated with swimming pool lessons is not considered as an anonymous noise source, and would most certainly not be a steady external source, as is demonstrated by the measured noise levels of swimming pool activity shown in Section 8.1 of the report. It is therefore considered inappropriate for these levels to be taken as stated. Should these levels be used, it would be expected that a discussion is provided to justify particular levels, but it is suggested that the actual criterion would be far lower than the levels referenced directly from BS 8233.

Finally, Section 8.3.2 presents "calculations and assessments are therefore to be carried out in order to satisfy the above requirements of BS8233: 2014". As noted above, BS 8233 is not appropriate for use in this manner. Should the author wish to provide some discussion and justification, then perhaps the reader can understand how the conclusion is drawn. As it is, the conclusion is meaningless. For example, if we apply the same crude -15 dB reduction for a partially open window to the skylight of the swimming pool itself, then the -7 dB below the criterion becomes 8 dB above the criterion. If the intent is to claim a level below the BS 8233 recommendations is "acceptable", then this simple inaccuracy renders the findings unacceptable.

1.6 L_{Amax} Assessment (Section 9 of the Report)

Section 9.1.1 notes that *"there is no legislation relating to criteria surrounding LAmax dB levels at facades during the day"*, which is agreed. The Report goes on to state *"the only criteria close to this would be from the World Health Organisation 'Guidelines for Community Noise' that suggests in order to not have a detrimental effect on sleep, night time periods from 23:00 – 07:00 hours should have no more than 10 events exceeding $_{LAmax}$ 45 dB. Whilst this can't be directly attributed to daytime levels, it does offer some level of acoustic value when establishing the likelihood of noise complaints from activities that could be deemed detrimental to neighbouring properties".*

How many "events" does the author believe to be occurring during each operational day of the swimming pool? If 10 events exceeding L_{Amax} 45 dB are unacceptable during night-time, how many events might be considered acceptable during daytime hours? Given that no discussion or justification is provided, this assessment becomes meaningless.

Section 9.1.2 is titled "Assessment Criterion", and states, *"The significance of the resulting noise on the residential property depends on the margin by which it exceeds the background noise levels. I consider the following to be appropriate for this assessment".* I do not consider this appropriate for the assessment. Again, no justification is provided.

Section 9.1.3 includes "Calculations". Again, no detail is provided here for the reader to understand how these levels have been established, which is severely lacking for a technical report. It is not clear whether the author has considered the partially open windows of the swimming pool. It is assumed that they have not, given that Section 4.3 discusses doors and windows of the swimming pool being closed. If we apply the same crude -15 dB reduction for a partially open window to the skylight of the swimming pool itself, then the calculated level of 40 dB becomes 55 dB, which is above the assessment level stated. While I do not agree that the assessment level is appropriate in the first instance, this demonstrates that the level calculated does not achieve the author's own criterion.

1.7 Additional Comments on Report

All assessments appear to be undertaken at the windows of the receptor properties. No assessment is provided for the external amenity areas, which are considered part of a residential dwelling where an occupant may choose to relax. As no assessments are provided to the neighbouring gardens, it is not clear whether or not an adverse impact might occur.

1.8 Additional Comments on the Rebuttal Letter Provided by Impact Acoustics

Impact Acoustics states that *“The EH Officer is ignoring the conclusions and facts of the noise report with the phrase 'Despite the NIA's conclusion', suggesting they're ignoring or even disagreeing with the conclusion of our assessment, yet they fail to offer any actual data that rebuts our conclusion”.* Given that the report is poorly constructed and there is a severe lack of justification or discussion throughout, it is not surprising that the EH Officer disagrees with the conclusions. It is not the job of the EH Officer to offer data to rebut the conclusions of the report – it is the job of the person who conducted the assessment and produced the report to provide a well-constructed and justified report, which they have failed to do in this case.

The mention of the Environmental Protection Act 1990 within the Impact Acoustics rebuttal letter is completely irrelevant to the situation at hand. The purpose of planning is to protect residents from adverse noise impacts. Had the report been robust enough to demonstrate that there are no adverse noise impacts, then there would be no need to mention this legislation.

The EH Officer stated, *“The operational hours proposed by the applicant are also of significant concern. The application requests hours of use that extend beyond typical daytime hours, including evenings and weekends, which would result in noise disturbance during times when residents are most likely to be in their homes and gardens. This is particularly problematic in a densely populated residential area like Allandale Avenue”.* Impact Acoustics responded, *“They are suggesting the activity of the swimming lessons ‘would result in noise disturbance during times when residents are most likely to be in their homes and gardens’. This simply isn't true and we have calculated and assessed the actual noise levels from a worst case scenario set of lessons and concluded a 'Low Impact' The council have failed to offer any technical data or calculations to suggest otherwise. Therefore, until they do, our assessment and conclusions remain accurate and a true representation of the proposed application”*.

It is not the job of the Local Authority to provide technical data or calculations for your assessment. Simply stating that your conclusions remain accurate and a true representation of the proposed application does not make it so. As detailed throughout this review, the report is severely lacking and is not considered robust enough such that proper conclusions can be drawn.